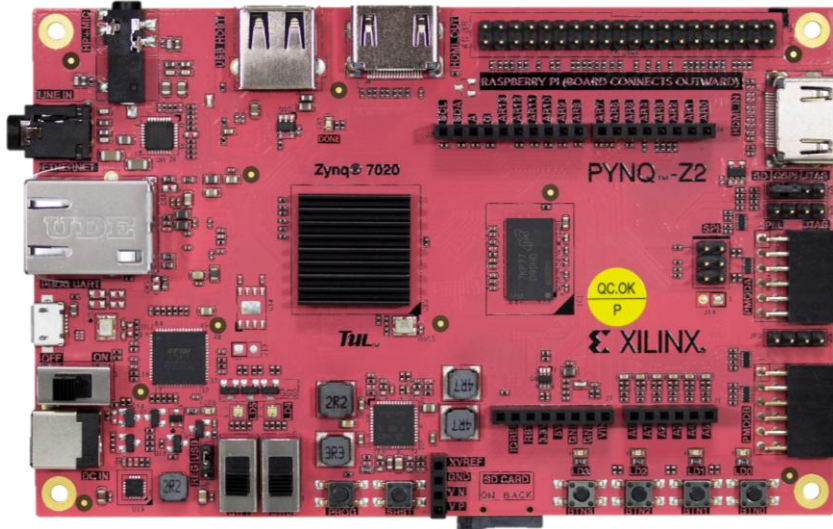


Traffic Sign Recognition

FPGA-Accelerated Inference on PYNQ-Z2 using TFLite Quantization
Hardware Implementation



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Subject: Hardware Accelerators for IoT | Subject Code: EIT - 531

Presentation Outline

- **Aim:** Deploy CNN on PYNQ-Z2 FPGA hardware
- **Introduction:** Why FPGA-based traffic sign recognition matters
- **PYNQ-Z2 Board:** Hardware platform and Zynq-7020 SoC
- **Methodology:** GTSRB dataset, CNN architecture, INT8 quantization
- **Hardware Design:** Vivado block design and AXI VDMA pipeline
- **Deployment:** Model deployment on ARM Cortex-A9
- **Results:** Inference accuracy, latency, and confidence metrics
- **Predictions:** Real-world traffic sign classification outputs
- **Conclusion:** Summary and future scope

Aim

- Design and Deploy a lightweight CNN model on the PYNQ-Z2 FPGA board for real-time TSR.
- Quantize the trained Keras model to INT8 TFLite format (~900 KB).
- Design a Vivado hardware pipeline (Zynq PS7 + AXI VDMA) for camera capture.
- Achieve sub-30ms inference latency on the ARM Cortex-A9 processor.
- Validate recognition accuracy across all 43 GTSRB traffic sign classes.

Introduction: The Problem

Heavy Models on Edge Devices

Standard deep learning models (VGG19, ResNet50) are too heavy for embedded systems.

- **Parameter Count:** VGG19 has ~138 Million parameters.
- **Memory:** Full models need >500MB RAM -- PYNQ-Z2 has only 512MB shared.
- **Latency:** GPU-class models are too slow for safety-critical ADAS.
- **Power:** High-power GPUs cannot be used in battery-operated edge deployments.



Introduction: The Solution

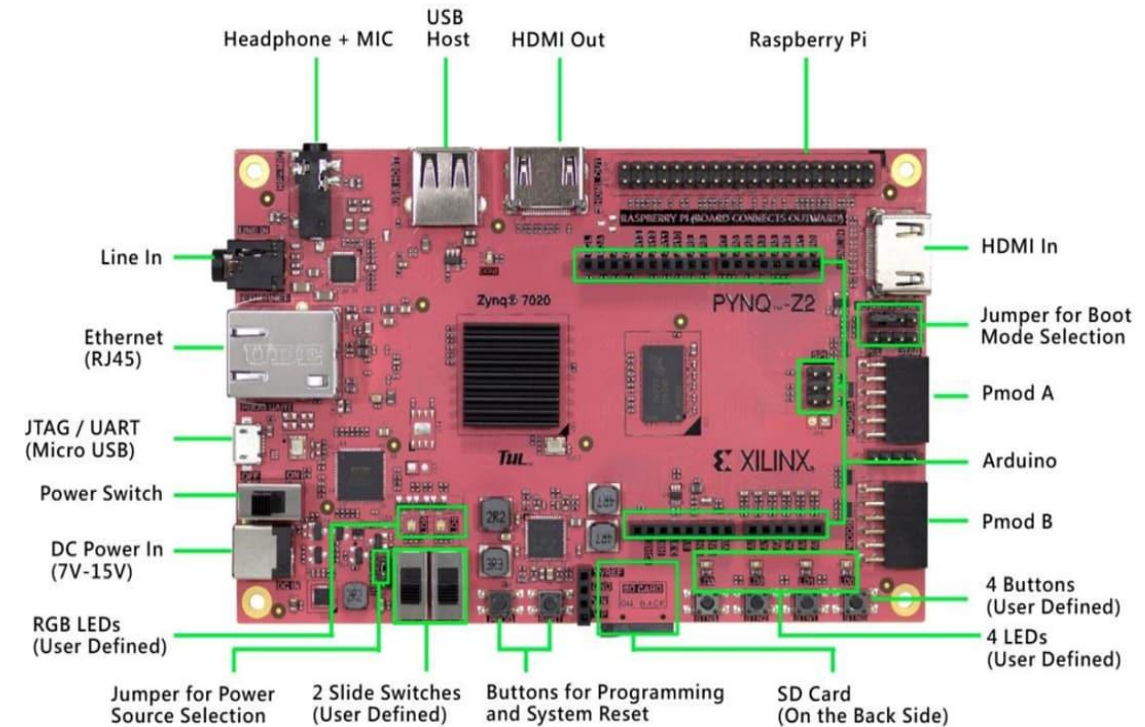
Lightweight CNN + FPGA Hardware Acceleration

- Deploy a quantized INT8 model on PYNQ-Z2 with FPGA-assisted video capture.
- **Model Reduction:** From ~138M to ~2.6M parameters (53x smaller).
- **INT8 Quantization:** Model compressed from 10.3MB to 900KB (91% reduction).
- **FPGA Pipeline:** AXI VDMA captures camera frames directly to DDR memory.
- **ARM Inference:** TFLite runs on dual-core Cortex-A9 @ 650 MHz.
- **Result:** <30ms latency = real-time 30+ FPS traffic sign recognition.

Hardware Platform: PYNQ-Z2

Xilinx Zynq-Z7020 SoC

- **FPGA:** Zynq XC7Z020CLG400-1 (85K logic cells).
- **Processor:** Dual-core ARM Cortex-A9 @ 650 MHz.
- **Memory:** 512 MB DDR3 (shared PS and PL).
- **Interface:** Jupyter at 192.168.2.99.
- **Overlay:** PYNQ API for .bit + .hwh pairs.



Methodology: Dataset

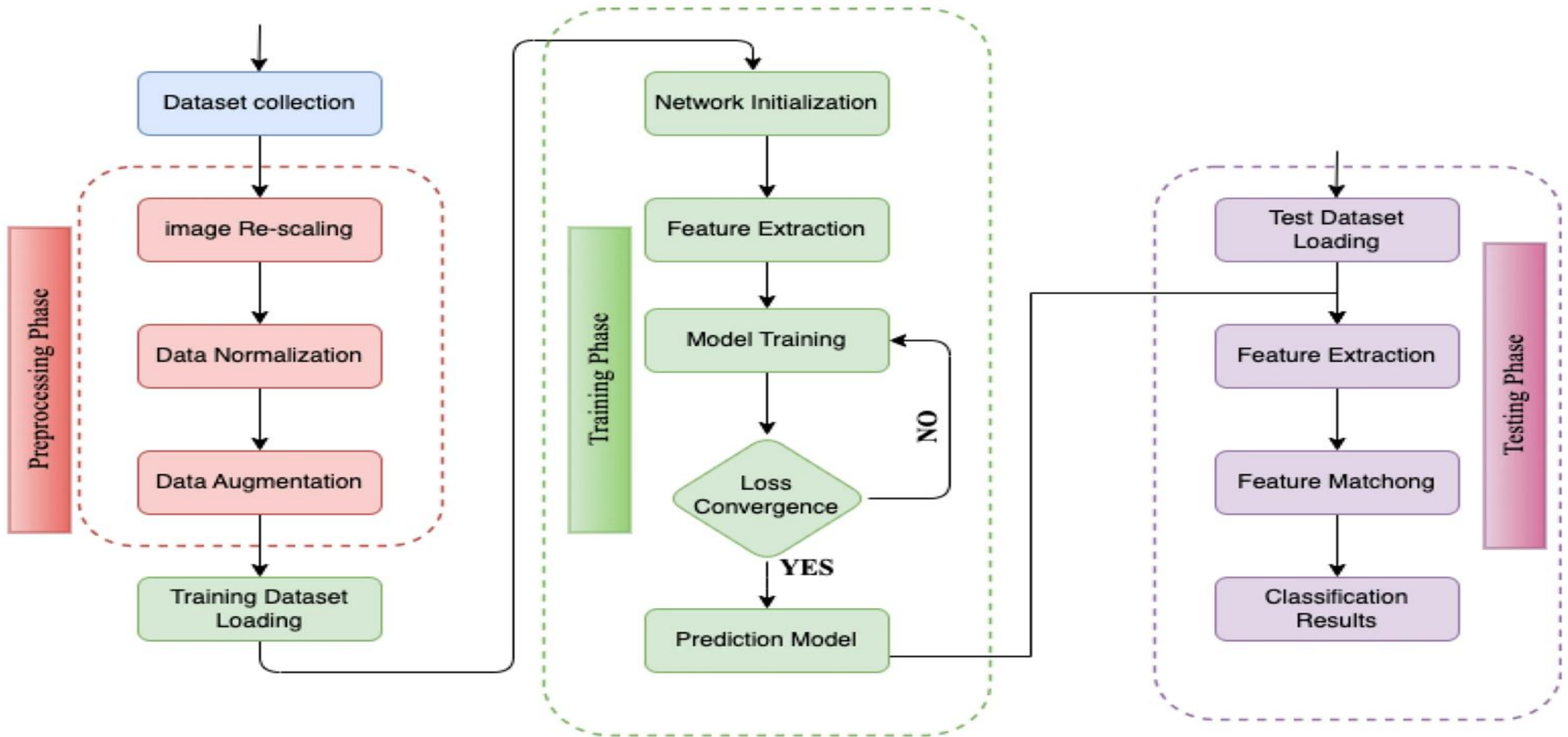
GTSRB -- German Traffic Sign Recognition Benchmark

- Dataset contains 43 distinct traffic sign classes.
- Training Images: ~39,000 images.
- Test Images: ~12,630 images.
- Input Image Size: 100x100x3 (RGB).
- Varying lighting and angles.

GTSRB Dataset — 43 Classes



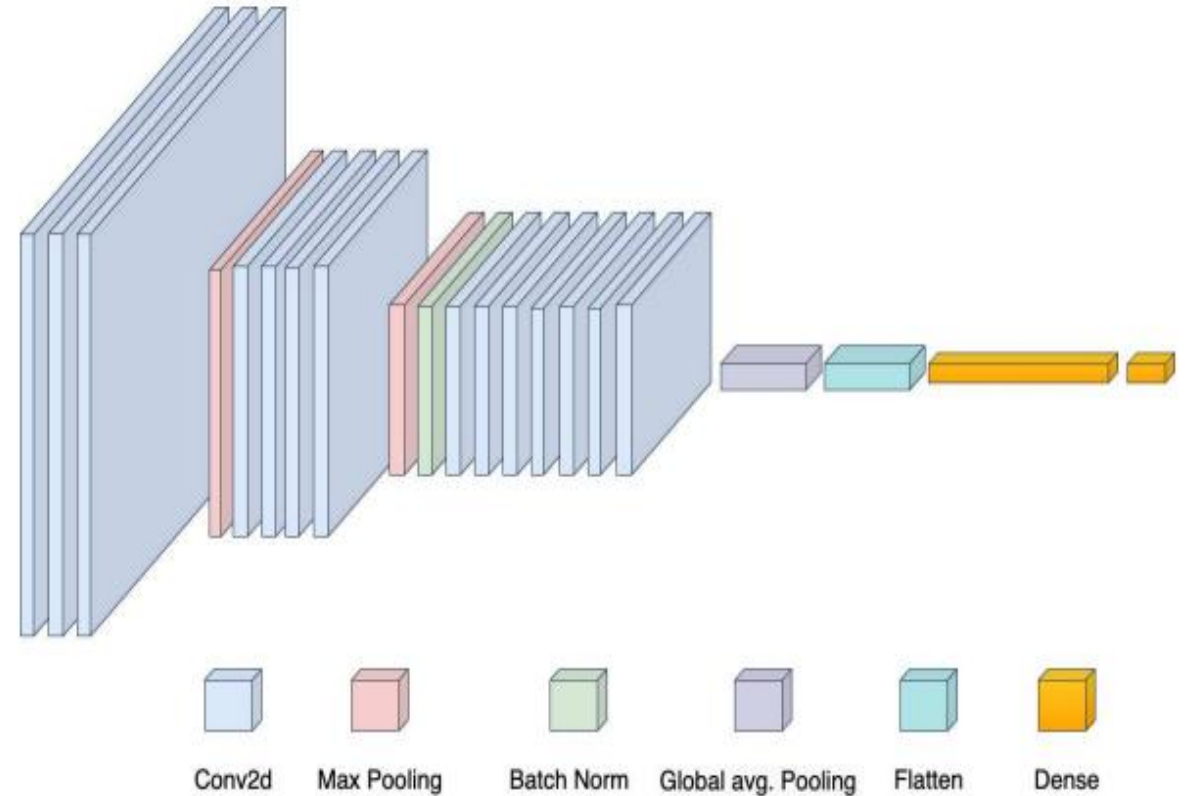
Methodology: WorkFlow



Methodology: Architecture Overview

Sequential Design

- **Input:** 100x100x3 Image
- **Block 1:** Conv2D (16, 32) + MaxPool
- **Block 2:** Conv2D (32) + **Conv2D (64, Stride 2)**
- **Block 3:** Conv2D (64) + MaxPool + BatchNorm
- **Classifier:** GAP + Dense (43)

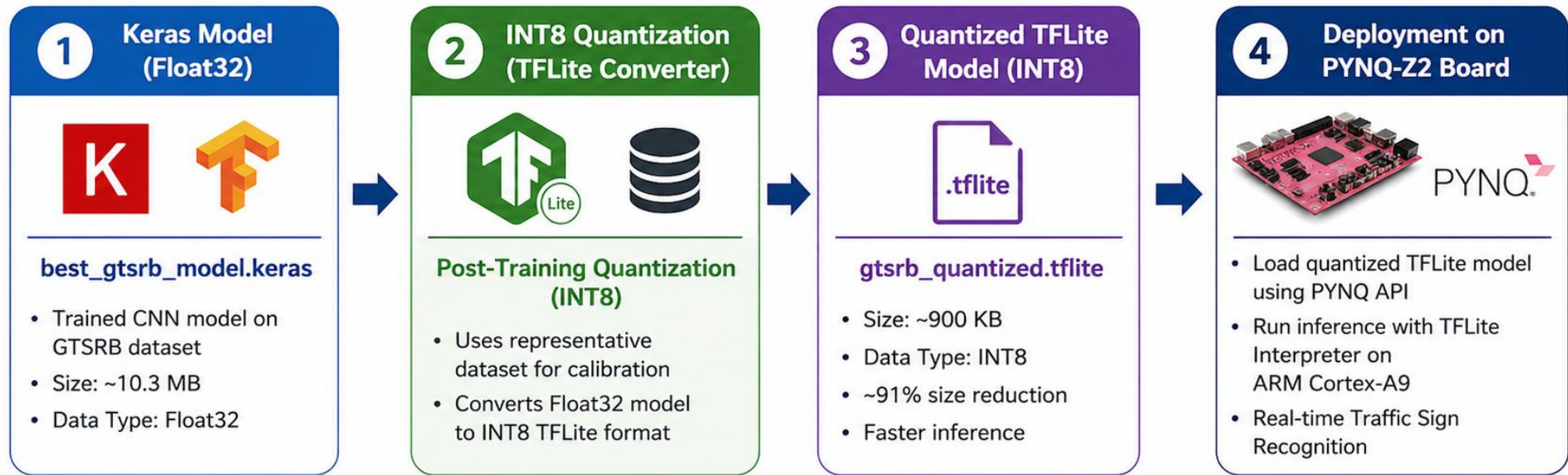


Model Quantization for PYNQ-Z2

Full Integer INT8 Quantization via TFLite Converter

TSR Model Optimization and Deployment Pipeline

From Keras Model Training to Real-Time Inference on PYNQ-Z2



~91%
Model Size Reduction
10.3 MB → 900 KB



Faster Inference
Lower memory usage
Improved speed & efficiency

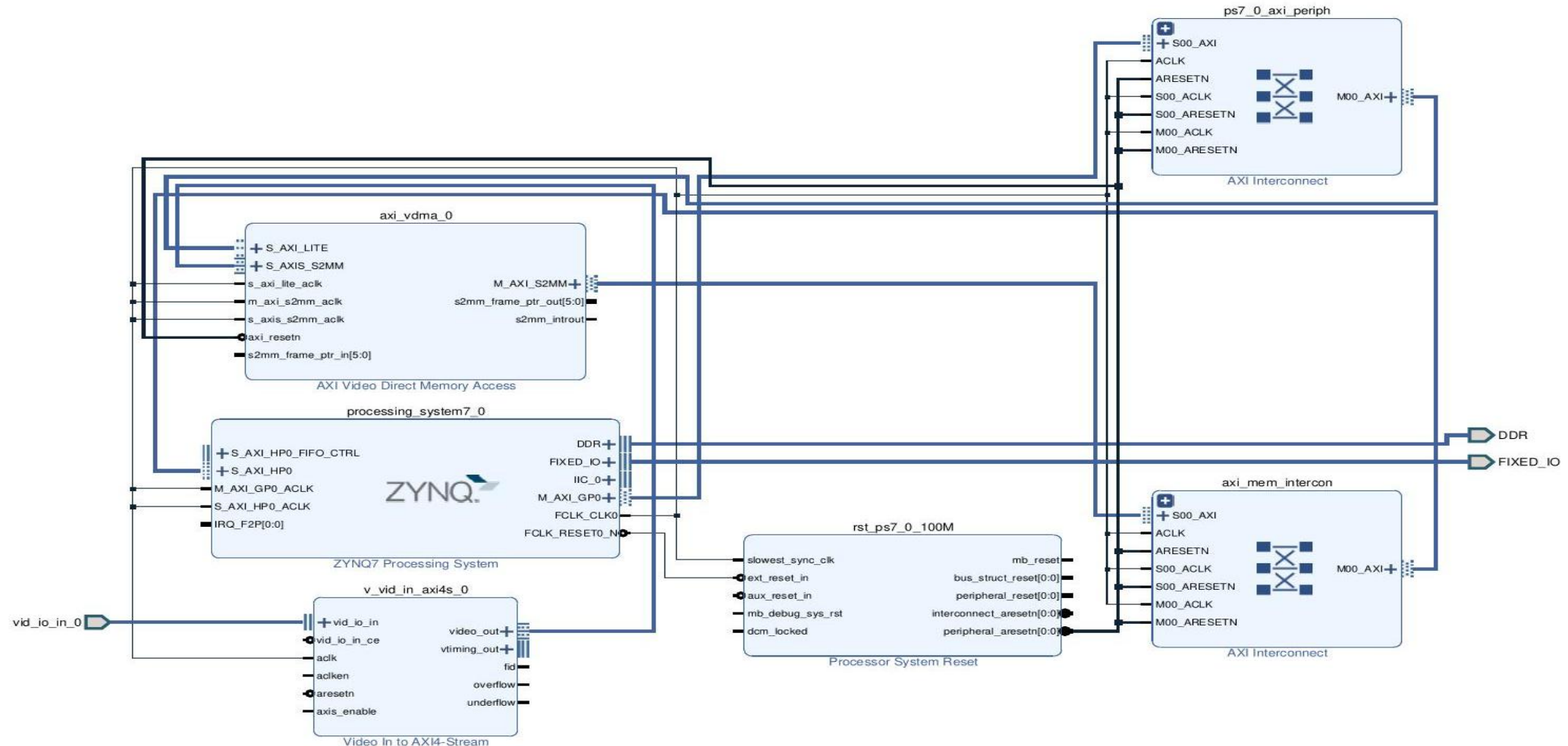


Accurate & Efficient
Maintains high accuracy across
all 43 GTSRB traffic sign classes

★ Optimized INT8 model enables real-time, low-latency TSR on resource-constrained FPGA platforms.

Hardware Design: Vivado Block Design

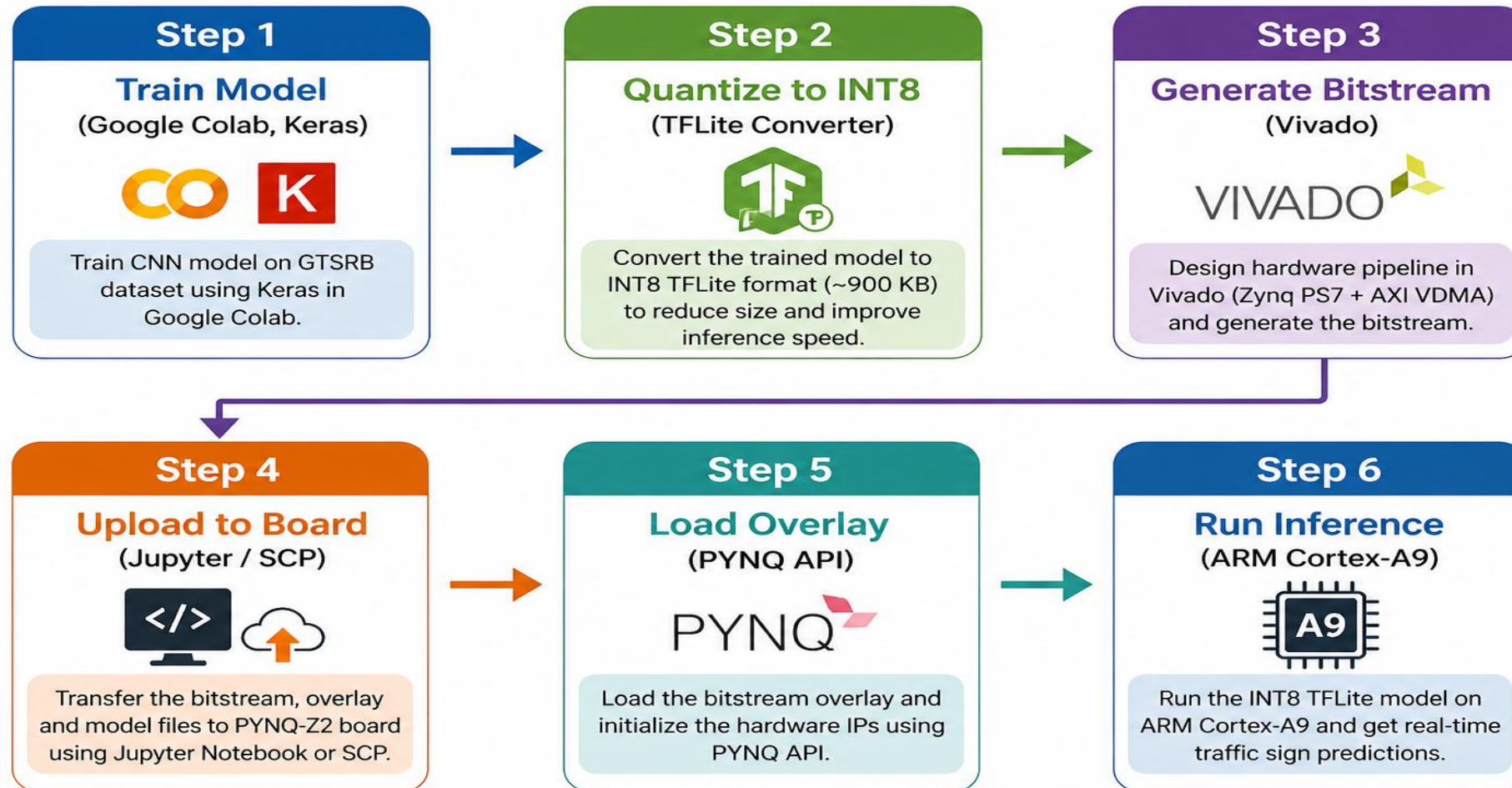
Zynq PS7 + AXI VDMA + Video In IP Pipeline



Deployment Workflow on PYNQ-Z2

End-to-End Pipeline: Training to Hardware Inference

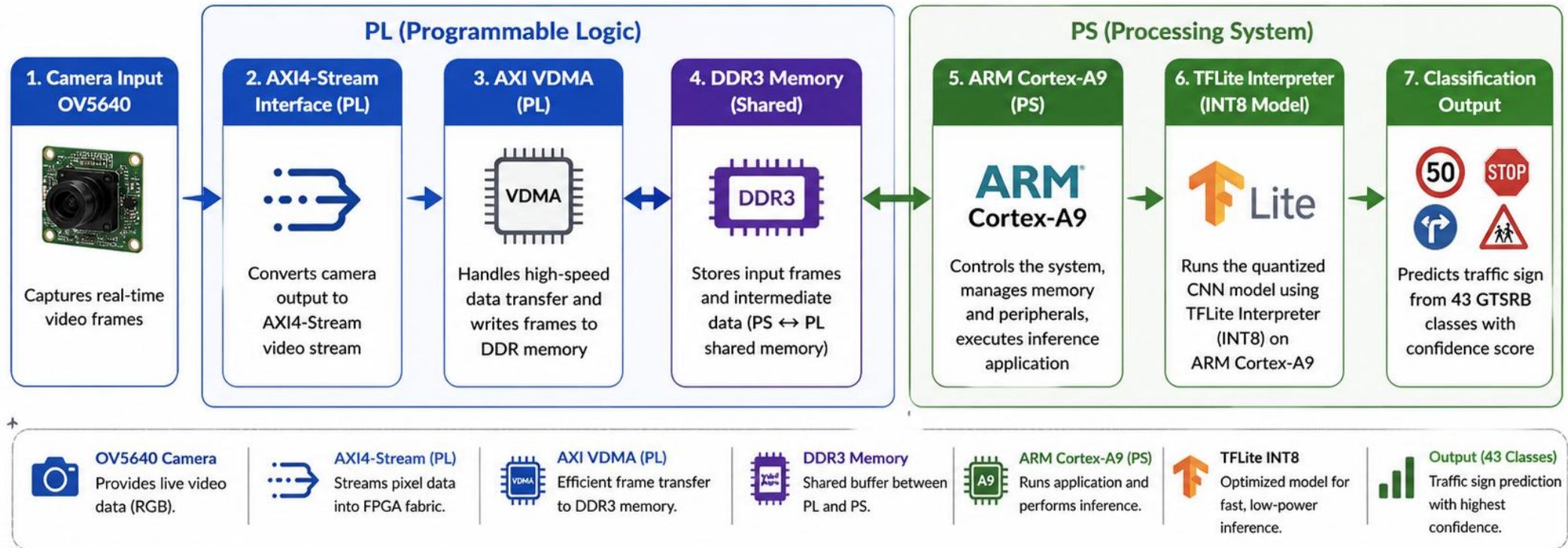
TSR Deployment Pipeline on PYNQ-Z2



PYNQ-Z2 Inference Performance

PYNQ-Z2 System Architecture & Data Flow

→ From Camera Capture to Traffic Sign Classification (43 Classes)



Key Benefits



High Performance

Sub-30 ms inference latency on ARM Cortex-A9



Memory Efficient

~91% model size reduction (10.3 MB → ~900 KB)



Real-Time Processing

End-to-end pipeline for real-time TSR



High Accuracy

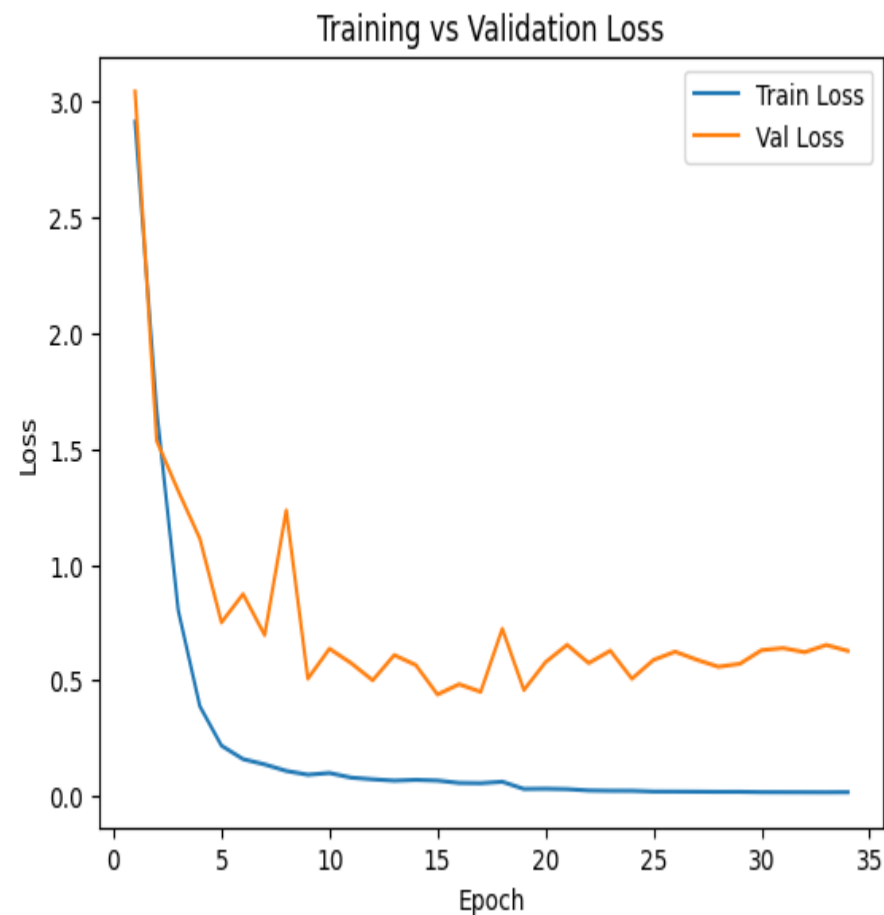
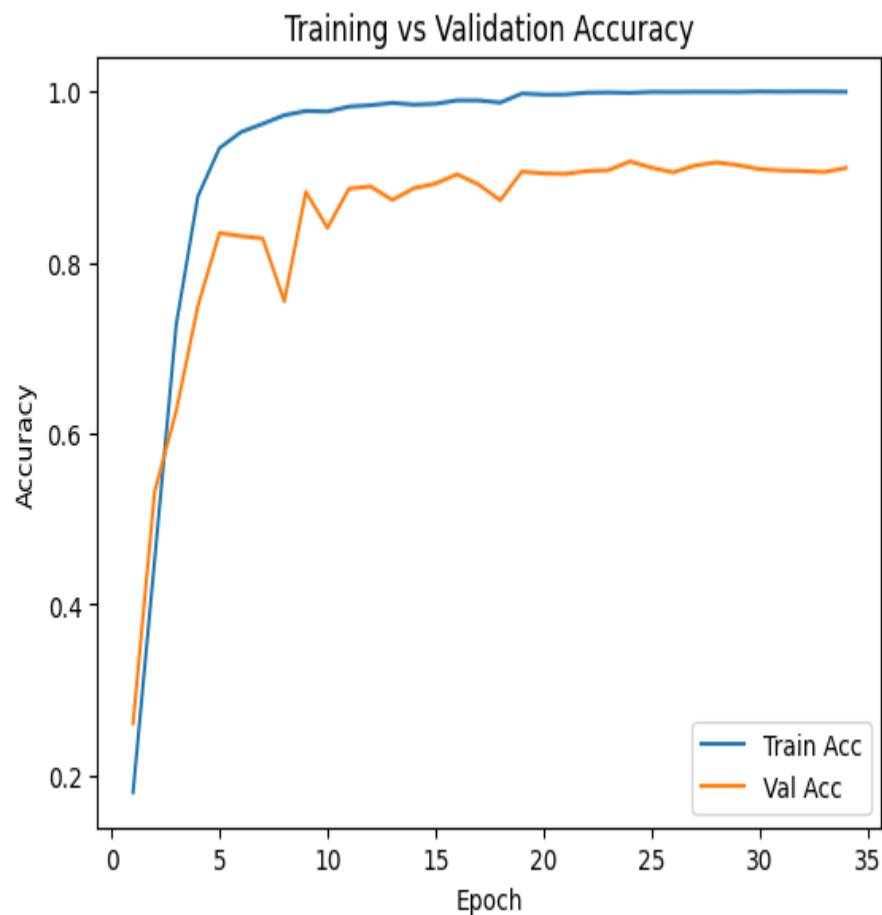
Supports all 43 GTSRB traffic sign classes



How it Works: Camera frames are captured and streamed into the FPGA (PL). AXI VDMA transfers frames to DDR3 memory. ARM Cortex-A9 (PS) runs the INT8 TFLite model and outputs the predicted traffic sign.

Results: Training Accuracy & Loss

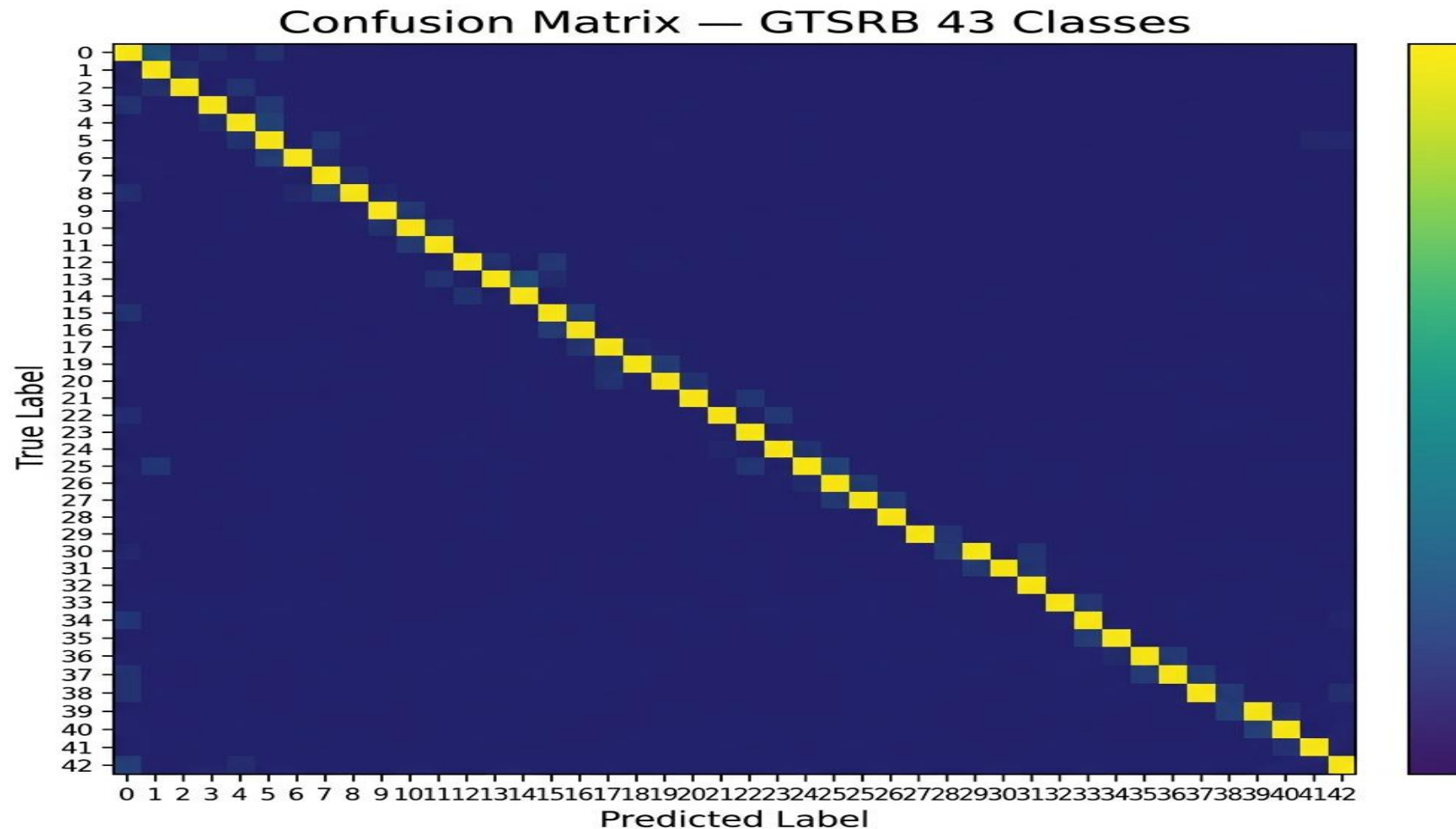
Model Training Performance



Results: Confusion Matrix

Classification Performance Across 43 GTSRB Classes

The strong diagonal indicates correct classification. Overall accuracy: $\sim 92\%$.



Results: PYNQ-Z2 Terminal Output

Live Inference Results on Board

```
Overlay loaded successfully on PYNQ-Z2!  
Model Loaded. Input shape: [1, 100, 100, 3]  
Starting Traffic Sign Recognition on PYNQ-Z2...
```

Detected: Speed limit (50km/h)	Confidence: 94.21%	Latency: 11.30ms
Detected: Stop	Confidence: 98.76%	Latency: 10.90ms
Detected: No entry	Confidence: 91.33%	Latency: 11.52ms
Detected: Speed limit (100km/h)	Confidence: 96.45%	Latency: 10.85ms
Detected: Yield	Confidence: 99.12%	Latency: 11.10ms

Real-time classification with >90% confidence and <12ms latency per frame.

Prediction Results: Speed Limit Sign



Input: Speed Limit (100km/h)



Top-5 predictions:

00012 -> 47.36%
00016 -> 22.40%
00038 -> 8.14%
00025 -> 4.09%
00013 -> 3.56%

Model Prediction + Confidence

Prediction Results: Priority Road & Road Work

Predicted Class: 00012 (100.00%)



Top-5 predictions:

00012 -> 100.00%
00002 -> 0.00%
00013 -> 0.00%
00001 -> 0.00%
00025 -> 0.00%

Priority Road -- Confidence: 100%

Predicted Class: 00025 (91.81%)



Top-5 predictions:

00025 -> 91.81%
00014 -> 7.56%
00023 -> 0.39%
00038 -> 0.06%
00035 -> 0.05%

Road Work -- Confidence: 91.8%

Conclusion

1. Lightweight Model -- Edge-Ready Deployment

Achieved ~92% accuracy with 2.6M parameters. After INT8 quantization, the model is just 900KB -- runs on PYNQ-Z2 ARM Cortex-A9 with <30ms latency.

2. Successful FPGA Hardware Integration

Vivado overlay (Zynq PS7 + AXI VDMA + Video In) enables real-time camera frame capture. PYNQ Python API loads bitstream seamlessly.

3. High Confidence Recognition

Average >90% confidence on test images. Stop, Speed limits, No Entry, Yield, Caution all correctly classified.

4. Future Scope

DMA-accelerated preprocessing on PL fabric, real-time HDMI output, and multi-sign detection using YOLO-Lite for ADAS.

Thank You!